

MPM1D – Exam Review – Units 1-2 Homework

1. Simplify each of the following by writing a single base and/or getting rid of all brackets, do not calculate.

a. $9^4 \times 9^5$ _____

b. $5^8 \times 5^4 \times 5$ _____

c. $3^9 \div 3^3$ _____

d. $7^8 \div 7^4 \div 7^0$ _____

e. $(2^8)^2$ _____

f. $(8^2)(8^4) \div 8^5$ _____

g. $7^6 \div 7^2 \times 7^5$ _____

h. $4^{12} \div (4^3)^3 \cdot 4^0$ _____

2. Simplify each of the following by writing as a single base, do not calculate.

a. $\left(\frac{2}{3}\right)^8 \left(\frac{2}{3}\right)^2 \div \left(\frac{2}{3}\right)^4$

b. $\frac{3^{16} \cdot (3^3)^3 \cdot 3}{3^{11} \times (3^2)^3}$

3. State the degree of each of the following polynomials.

a. $12x^2 y^4 z^{14}$

b. $13ab + 12cd - 2a^2 b^3$

c. $8 + y - 3m + 17y^4 - 8m^2n + 4ab$

4. Identify each of the following polynomials as a monomial or binomial or trinomial.

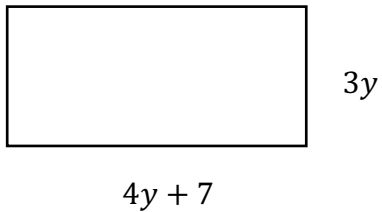
a. $3a^2b - 7b^2c - 18$

b. $4c - 8a$

c. $24a^3b^4c^2d$

5. Write a simplified expression for the perimeter and a simplified expression for the area of the rectangle below. Show your work.

Recall that $P = \Sigma sides$ and $A = LW$



6. Simplify each expression below. Show your work.

a. $6x + 9y + 8y - 5x$

b. $12a^2b - 9ab^3 + 17a^2b - 7ab^3 + 9ab^2$

c. $27 - 5a - 14b^2 + 41 + 7a - 19b^2 + 13a$

d. $5(y + 4)$

e. $(4y - 7) - (y - 8) - (-7y + 3)$

f. $2y(7y^2 - 3y + 4)$

g. $5y^3(2y^2 - 6ay)$

h. $-2y^2(-5y^5 + 7y - 4a + y^4)$

i. $2y(4 - 8y) - 3(2y^2 + 6)$

j. $\frac{-(3m^2n^5)(6m^5n^4)}{(2m^2n)^3}$

Unit 2 – Equations

1. Solve each equation below and check your answer.

a. $\frac{x}{7} = 21$

b. $4x - 15 = 13$

c. $9 + \frac{x}{5} = 3$

d. $4(x - 6) = -52$

e. $9 = \frac{x-8}{2}$

f. $3x - 8 = -2x + 32$

g. $\frac{x}{4} = \frac{x-5}{2}$

h. $7 = \frac{3x+2}{5}$

2. Solve for the unknown variable and check your answer.

$$\frac{3}{4}(x + 2) = \frac{2}{3}(x - 5)$$

3. Write an expression for each one of the following statements. Let x be the unknown number.

- a. Nineteen more than a number _____
- b. One-quarter of a number decreased by three _____
- c. Seven less than twice a number _____
- d. The sum of five a number is divided by four _____

4. Isolate for A in each of the following equations. Show your work.

a. $AB = C$

b. $AD + B = C$

c. $\frac{A^2}{B} = C$

d. $\frac{A}{C} - B = D$

e. $\sqrt{A} - B = C$

f. $\frac{BC}{A} = D$

5. Write an expression for each description below. Don't forget to include let statements to describe your variables.

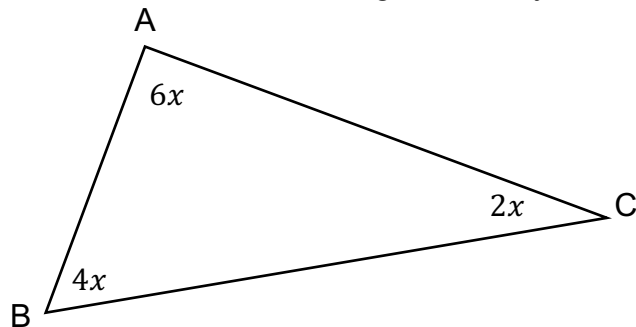
a. On a Math Contest, students receive 7 marks for each correct answer and lose 3 marks for each incorrect answer.

b. A team plays 50 games in a season. They can either win or lose, but there are no ties. The team earns 3 points for every win and loses 1 point for every loss.

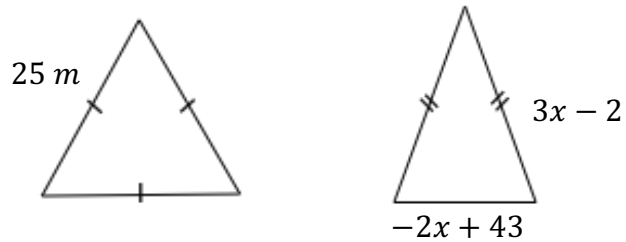
6. Mr. T works in a clothing store in the summer. He earns \$130 per day and a bonus of \$7 for every shirt that he sells and a bonus of \$9 for every pair of pants that he sells. Let E be his total earnings (in \$), let x be the number of shirts he sells, and let y be the number of pairs of pants he sells.

- Write an equation to show Mr. T's earnings in one day.
- Use your equation to determine how much Mr. T would earn if he sold 5 pairs of pants and 6 shirts in one day. Show your work.

7. Solve for the measure of each angle. Show your work.



8. The following two triangles have the same perimeter in metres. Determine the dimensions of the three side lengths of the second triangle.



For each one of the following word problems, please complete the following:

- i. Write let statements with one variable
- ii. Write an equation
- iii. Solve the equation
- iv. Check your solution
- v. Write a therefore statement to answer the question

9. River is five years younger than Mountain. If the sum of their ages is 51, how old is each individual?

10. April and June had a bouncing contest. If April bounced seven less times than twice the bounces of June, and their combined bounces totaled 383, how many times did each girl bounce?

11. Three angles are supplementary. The second angle is twice the measure of the first angle. The third angle is 6° less than triple the first angle. Determine the measure of each of the three angles.

12. The sum of four consecutive **even** integers is 268. Determine the value of the integers.

Answer Key – Unit 1

1a) 9^9

1b) 5^{13}

1c) 3^6

1d) 7^4

1e) 2^{16}

1f) $8^1 \rightarrow 8$

1g) 7^9

1h) 4^3

2a) $\left(\frac{2}{3}\right)^6$

2b) 3^9

3a) 20

3b) 5

3c) 4

4a) trinomial

4b) binomial

4c) monomial

5) $P = 14y + 14 \text{ units}$

$A = 12y^2 + 21y \text{ units}^2$

6a) $x + 17y$

6b) $29a^2b + 9ab^2 - 16ab^3$

6c) $68 + 15a - 33b^2$

6d) $5y + 20$

6e) $10y - 2$

6f) $14y^3 - 6y^2 + 8y$

6g) $10y^5 - 30ay^4$

6h) $10y^7 - 14y^3 + 8ay^2 - 2y^6$

6i) $-22y^2 + 8y - 18$

6j) $-\frac{9mn^6}{4}$

Answer Key – Unit 2

1a) $x = 147$

1b) $x = 7$

1c) $x = -30$

1d) $x = -7$

1e) $x = 26$

1f) $x = 8$

1g) $x = 10$

1h) $x = 11$

2) $x = -58$

3a) $x + 19$

3b) $\frac{x}{4} - 3$

3c) $2x - 7$

3d) $\frac{5+x}{4}$

4a) $A = \frac{C}{B}$

4b) $A = \frac{C-B}{D}$

4c) $A = \sqrt{BC}$

4d) $A = BC + DC$

4e) $A = (B + C)^2$

4f) $A = \frac{BC}{D}$

5a) $T = 7x - 3y$

Let T be the total number of points

Let x be the number of correct answers

Let y be the number of incorrect answers

5b) $T = 3x - y$

Let T be the total number of points

Let x be the number of wins

Let y be the number of losses

6a) $E = 130 + 7x + 9y$

6b) \$217

7) $30^\circ, 60^\circ, 90^\circ$

9) 25 m

10) Mountain is 28 years old

River is 23 years old

11) June bounced 130 times

April bounced 253 times

12) $31^\circ, 62^\circ, 87^\circ$

13) 64, 66, 68, 70